Quiz 1

1- In Display, The red, green, and blue components of a pixel each use a pair of bits. These pairs are fed to three digital-to-analog converters (DAC's), which
a. choose the color
b. convert logical values into actual voltages
c. converts analog to digital signal

2- Which of the following is a primary advantage of using scalable vector graphics (SVG)?
a. It is limited to black and white graphics.
b. It is resolution-independent and scales without loss of quality.
c. It cannot be manipulated with CSS or JavaScript.
d. It only supports raster images.

3- Frame Buffer is :
a. A portion of flash-memory-register containing a bitmap that drives a video display.
b. A portion of Hard-disk containing a bitmap that drives a video display.
c. A portion of read-only-memory (ROM) containing a bitmap that drives a video display.
d. A portion of random-access-memory (RAM) containing a bitmap that drives a video display.

4- ——— Used for less details Images such as logos
a. Vector display
b. none
c. Raster display

5- Which of the following is a characteristic of vector graphics?
a. Always larger in file size than raster images
b. Composed of pixels
c. Made up of paths defined by mathematical equations
d. Limited to a fixed resolution

6- ————analyze pictures to derive descriptions
a. Image Processing
b. computer graphics
c. all
d. computer Vision

7- Using the direct coding method for Pixel color representation, How much memory is required for a 4-bit/pixel frame buffer of size 512x1024?
a. 4M byte
b. 256K byte
c. 2M byte
d. 512K byte

8- If a video is recorded at 60 FPS, but played back on a 30 Hz monitor, how many frames will be shown per second?
a. 30 FPS
b. 15 FPS
c. 120 FPS
d. 60 FPS

9- -------describes how objects appears (it defines attributes that describe color texture or shape of the object
a. all mentioned
b. mathematical model
c. physical model
d. Geometric model

10- Can we find a Raster Display in the real life?.
a. True.
b. False.

11- Graphics pipeline Defined as : It is a ..... that describes what steps a graphics system needs to perform to render a ..... to a .....
a. Design model - 3D scene - 2D screen
b. conceptual model - 3D scene - 2D screen
c. Mathematical model - 3D scene - 2D screen
d. conceptual model - 2D scene - 3D screen

12- Bitmap is :
a. Grid of pixels.
b. Grid of lines.
c. Grid of edges.
d. Grid of points.

13- POLYLINES is :
a. connected sequence of angles.
b. connected sequence of shapes.
c. connected sequence of straight lines.

14- A display that consists of a regular grid of pixels and is normally driven from a frame buffer called :
a. Raster Display.
b. LCD Display.
c. Vector Display.
d. LED Display.

15- Elements of Pictures Created in Computer Graphics
a. raster Image
b. polyline
c. all of the above
d. Text

16- What is a common characteristic of interactive graphics?
a. Static images displayed sequentially
b. Real-time rendering and user feedback
c. Use of traditional print media
d. Pre-rendered animations

17- Using the Look-Up Tables (LUTs) method for Pixel color representation, How much memory is required for a 4-bit/pixel frame buffer of size 512x1024 along with LUT of 6 byte wide
a.1M byte
b.4K byte
c.262144 byte
d.262240 byte.

18- Which of the following is an example of a non-interactive graphics system?
a. CAD software for 3D modeling
b. Video game
c. Virtual reality simulation
d. Animated movie

19- The instruction glutInitWindowPosition(100, 150); will apply x, and y coordinate from the origin pixel of display from:
a. right – up
b. right - down.
c. left - down.
d. left - up.

20- Concave Shape is:

a. Shape that have start point and end point not connected.

b. Polygon Shape Which have number of edges equal to number of angles.

c. Polygon Shape determined if a straight line can be drawn between two points in the shape that lies outside the boundary of the shape.

d. Polygon Shape determined if a straight line can be drawn between two points in the shape that lies inside the boundary of the shape.

Quiz 2

1- clipping window is the viewport?
a. True
b. False

2- myReshape(GLsizei W, GLsizei H); function is used to----?
a. none
b. resize the view window
c. both display and view window
d. resize the display window

3- viewing port insure that the image will be displayed?
a. False
b. true

4- for the following setting{glOrtho2D(0,2,1,1) and glViewport(40,60,360,240)} the triangle of the vertex (10,10,10,10), (20,20,20),(30,30,30)
a. the triangle is clipped
b. the triangle is displayed
c. nothing appeared

5- for the following setting{glOrtho2D(0,2,1,1) and glViewport(40,60,360,240)} the image is displayed on ---?
a. (0,2,1,1) dimension
b. mapped to (0,2,1,1) dimension
c. mapped to (40,60,360,240) dimension
d. (40,60,360,240) dimension

6- the number of viewports(of dimension (10,20,50,40)) that can be displayed in a window size(40,100) . the viewports non overlapped and window positions (10,10)?
a. 40
b. 27
c. 5
d. non

7- ----------is where to be displayed?
a.Window
b.all
c.graphics
d.Viewport

8- The mapping of a two-dimensional, world-coordinate scene description to device coordinates is called ?
a.viewing transformation
b.display window
c.clipping window.
d.Mapped Normalized coordinates

9- If the user makes the screen window bigger the previous viewport still be viewed as previous without any change with resizing ?
a. False
b. True

10- to display the image in a short window the aspect ratio is-----?
a. Change the window size
b. Less than 1
c. Equal 1
d. Greater than 1

11- to display the image of dimension (30,30) in a display window five times on window diagonals ----?
a. For (int i=0,i<5,i++) glViewport(i,i,i*30,i*30)
b. non
c. For (int i=0,i<5,i++) glViewport(i,i,30+30*i,30+30*i)
d. For (int i=0,i<5,i++) glViewport(i,i,i+30,i+30)

12- to flip the window around the x axis you set the window of dimension 640X480 to-----?
a. setwindow(0,640,480,0)
b. setwindow(0,640,0,480)
c. setwindow((640,0,0,480)
d. setwindow(640,0,480,0)

13- to display the image in a tall window the aspect ratio is-----?
a. Equal 1
b. Change the window size
c. Less than 1
d. Greater than 1

14- to flip the window around the y axis you set the window of dimension 640X480 to-----?
a. setwindow((640,0,0,480)
b. setwindow(640,0,480,0)
c. setwindow(0,640,480,0)
d. setwindow(0,640,0,480)

15- the image you draw during the run an empty screen appears. This error is due to---?
a.all
b.viewing window size
c.glorthognal setting
d.display window size

16- we could set up a separate two-dimensional, viewing-coordinate reference frame for specifying the?
a.viewing transformation
b.clipping window.
c.Mapped Normalized coordinates
d.display window

17- ---------- is What to be displayed?
a. all
b. graphics
c. Viewport
d. Window

18- If the user makes the screen window bigger the previous viewport could still be viewed as previous with resizing ?
a.False
b.True

Quiz 3

1- From the disadvantage of DDA
a. 3_We cannot use the multiplication method.
b. 2 and 3
c. 1 and 2
d. 1_The floating-point arithmetic is time-consuming
e. 2_The accumulation of round-off error in successive additions of the floating-point increment.

2- however, the jaggies(stair step appearance) will be apparent to some extent even at the highest resolution possible with current technology, the jaggies(stair step appearance) will be apparent to some extent
a. false
b. true

3- In which algorithm is the decision parameter updated during the line drawing process?
a. Neither algorithm
b. Both algorithms
c. Bresenham's Midpoint Algorithm
d. Digital Differential Analyzer (DDA)

4- if the slop of line is small negative , we decide to change or not each time the value of -------
a. non
b. X value
c. Y value
d. decide each time step which to increment

5- The main advantage of Bresenham's algorithm over DDA is:
a.It is easier to implement
b.It can handle all types of lines equally well
c.It produces higher-quality lines
d.It requires less computational overhead

6- In Bresenham's algorithm, the decision parameter (P) is used to determine:
a. Whether to increment y or not
b. The slope of the line
c. The length of the line
d. The color of the line

7- In Bresenham's algorithm, when the decision parameter (P) is negative, the next pixel:
a. Is diagonally up and to the right
b. Is directly to the right
c. Is directly above
d. Is diagonally down and to the right

8- Which of the following equations is used to update the decision parameter in Bresenham's algorithm if the decision parameter value is less than zero?
a. P_new = P + Δx - Δy
b. P_new = P + 2Δy
c. P_new = P + 2Δx
d. P_new = P + 2Δy - 2Δx

9- Which of the following is an advantage of the DDA over other line drawing algorithms?
a. It uses integer arithmetic exclusively
b. It requires less memory
c. It provides better quality lines
d. It is simpler to implement

10- Which of the following equations is used to update the decision parameter in Bresenham's algorithm if the decision parameter value was greater than zero?
a. P_new = P + 2Δy - 2Δx
b. P_new = P + 2Δy
c. P_new = P + Δx - Δy
d. P_new = P + 2Δx

11- The primary advantage of Bresenham's algorithm over other line drawing algorithms is:
a. It requires more memory
b. It uses floating-point arithmetic
c. It is faster and uses only integer calculations
d. It can draw curves easily

12- DDA Line Drawing Algorithm produce the following points for a line with endpoints (5,7) and (8,14).
a. (5,7) (5,8) (6,10)
b. (7,11) (7,12) (7,13)
c. (5,9) (6,10) (6,11)

13- What is the primary purpose of the Digital Differential Analyzer?
a. To manage memory allocation
b. To generate and display curves and lines
c. To perform arithmetic operations
d. To store large datasets

14- In the DDA algorithm, what is the significance of the variable m?
a. It determines the color of the line
b. It is used to calculate the length of the line
c. It represents the midpoint of a line segment
d. It indicates the slope of the line

15- Which of the following equations is used to calculate the first pixel position between endpoints (7,8) and (3,2)in DDA?
a. (x+1,y+m)
b. (x-1,y-m)
c. (x-(1/m),y-1)
d. (x+(1/m),y+1)
e. (x+1,y-1)

16- What is the initial value of the decision parameter (P) when drawing a line using Bresenham's algorithm with endpoints (20,10) and (30, 18)?
a. P = 6
b. P = 16
c. P = 4
d. P = 0

Quiz 4

1- By using Mid point algorithm to draw a circle with a radius of 9 units, along the circle octant in the 1st quadrant includes
a. (1,10), (2,9), (3,9), (4,9),(5,9),(8,8),(7,7)
b. (1,9), (2,10), (3,11), (4,11),(5,11),(8,12),(7,13)
c. (1,9), (2,9), (3,9), (4,9),(5,9),(8,8),(7,7)

2- If the points of an ellipse that lies in the first quad are (1,6), (2,6), (3,6), and(4,5) then the similar points on the second quad are:
a. (-1,6), (-2,6), (-3,6), (-4,5)
b. (-1,-6), (-2,-6), (-3,-6),(-4,-5)
c. (1,-6), (2,-6), (3,-6),(4,-5)

3- If the world coordinates of a point are (xW, yW) and the corresponding viewport coordinates are (xV, yV), what is the basic formula to convert from world coordinates to viewport coordinates?
a. XV= $XVmin+((XVmax-XVmin)/(XWmax-XWmin)* (XW-XWmin))$
b. XV=$(XVmax-XVmin)/(XWmax-XWmin)* (XW-XWmin)$
c. xv=xw/2
d. XV= $XVmin+(XVmax-XVmin)/(XWmax-XWmin)$

4- In the Midpoint Ellipse Algorithm, the two axes of the ellipse are defined by which of the following parameters?
a. Radius and Diameter
b. Height and Width
c. Major and Minor Axes
d. X and Y Coordinates

5- there is no limit to how big we can make the frame buffer and still maintain the refresh rate at 60 frames or more per second
a. false
b. true

6- What are the main parameters needed to implement the Midpoint Ellipse Algorithm?
a. The four corners of the ellipse
b. Radius and angle
c. Only the center point
d. Center coordinates and the lengths of the major and minor axes

7- What is the initial decision parameter used in the Midpoint Circle Drawing Algorithm for the first point of the circle?
a. p=r*r
b. p=r
c. p=1-r
d. p=r-2

8- What is the primary purpose of antialiasing in computer graphics?
a.To enhance texture details
b.To increase rendering speed
c.To reduce color depth
d.To minimize the visual distortion of edges

9- What is the purpose of the decision parameter in the Midpoint Ellipse Algorithm?
a. To decide which pixel to plot next
b. To determine the color of the ellipse
c. To set the rendering speed
d. To calculate the area of the ellipse

10- Which of the following techniques is commonly used for antialiasing?
a.Depth buffering
b.Texture mapping
c.Supersampling
d.Rasterization

11- ----------to model light and to perform simulations for animation
a.mathematical model
b.physical model
c.Geometric model
d.all mentioned

12- Bresenham Line Drawing for a line line with endpoints (10, 0) and (20, 8). This line has a slope of 0.8, includes
a.(12,2), (13,2), (14,3), (15,4)
b.(12,2), (13,3), (14,3), (15,4)
c.(12,2), (13,3), (14,4), (15,4)

13- Using the mid-point circle drawing algorithm, what's the initial point used to draw a circle with a radius of 8 units, along the circle octant in the 1st quadrant
a.(8,8)
b.(0,8)
c.(0,0)
d.(8,0)

M-T Mack up

1- if the points of a circle which lies in the first octant are (5,11),(7,12),(9,12) then the similar points on third octant are:
a. (-11,5) (-12,7) (-12,9)
b. (-5,-11),(-7,-12),(-9,-12)
c. (11,5) (12,7) (12,9)
d. (5,-11),(7,-12),(9,-12)

2- the number of viewports(of dimension (10,20,50,40)) that can be displayed in a window size(40,100) . the viewports non overlapped and window positions (10,10)?
a. 5
b. 27
c. 40
d. non

3- the DDA algorithm primarily operates in which type of coordinate system?
a. Cartesian coordinates
b. Polar coordinates
c. Spherical coordinates
d. Cylindrical coordinates

4- myReshape(GLsizei W, GLsizei H); function is used to----?
a. resize the view window
b. both display and view window
c. resize the display window
d. none

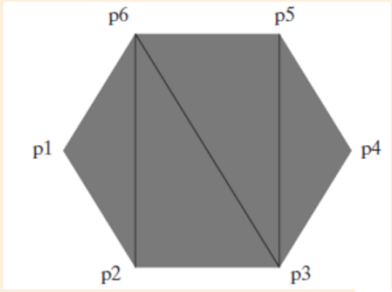
5- The mapping of a two-dimensional, world-coordinate scene description to device coordinates is called ?
a. display window
b. clipping window.
c. Mapped Normalized coordinates
d. viewing transformation

6- Which algorithm is known for using only integer arithmetic in line drawing?
a. Digital Differential Analyzer (DDA)
b. Neither algorithm
c. Bresenham's Midpoint Algorithm
d. Both algorithms

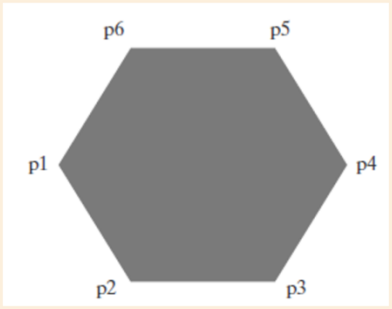
7- connect the right figure

Your answer is incorrect.

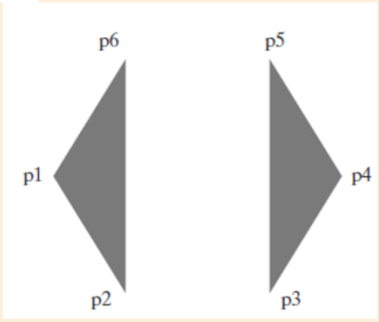
The correct answer is:



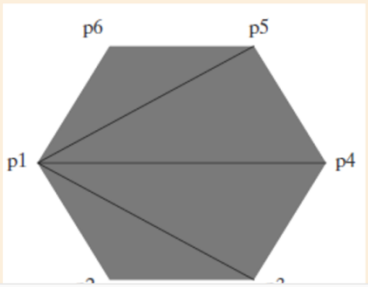
→ GL TRIANGLE STRIP,



→ GL POLYGON,



→ GL TRIANGLES,



→ GL TRIANGLE FAN.

8- for the following setting{glOrtho2D(0,2,1,1) and glViewport(40,60,360,240)} the image is displayed on ---?

a. mapped to (0,2,1,1) dimension
b. mapped to (40,60,360,240) dimension
c. (0,2,1,1) dimension
d. (40,60,360,240) dimension

9- to display the image of dimension (30,30) in a display window five times on window diagonals ----?

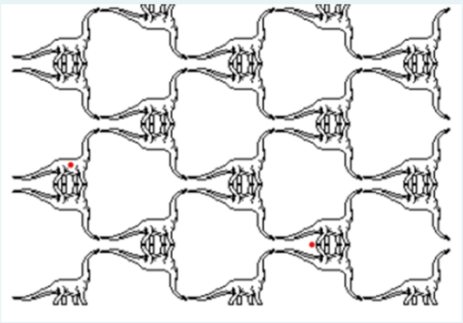
a. non
b. For (int i=0,i<5,i++) glViewport(i,i,i+30,i+30)
c. For (int i=0,i<5,i++) glViewport(i,i,i*30,i*30)
d. For (int i=0,i<5,i++) glViewport(i,i,30+30*i,30+30*i)

10- the flowing code draw

the flowing code draw

```
setWindow(0, 640.0, 0, 480.0);
for(int i = 0; i < 5; i++)
   for(int j = 0; j < 5; j++)
   {
      glViewport(i * 64, j * 44, 64, 44);
      drawPolylineFile("dino.dat");
   }
```

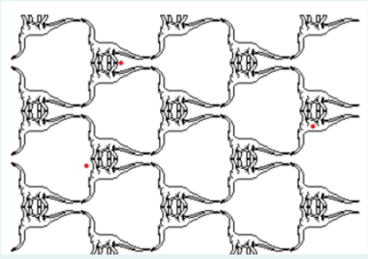
a.



b.



c.



Your answer is incorrect.

The correct answer is:



11-

for midpoint circle drawing if $p_0 = 1 - r$

>0-new condition is ----------else ----------

a. non

b. $p_{k+1} = p_k + 2x_{k+1} + 1$,---$p_{k+1} = p_k + 2x_{k+1} + 1 - 2y_{k+1}$

c. $p_{k+1} = p_k + 2x_{k+1} + 1 - 2y_{k+1}$, $p_{k+1} = p_k + 2x_{k+1} + 1 - 2y_{k+1}$

Your answer is incorrect.

The correct answer is: $p_{k+1} = p_k + 2x_{k+1} + 1 - 2y_{k+1}$, $p_{k+1} = p_k + 2x_{k+1} + 1 - 2y_{k+1}$

12- _DDA Line Drawing for a line with endpoints (1,5) and (10,9).

a. (1,5) (2,5) (3,6) (4,6)
b. (1,5) (2,5) (3,6) (3,7)
c. (1,5) (2,6) (3,7) (4,7)

13- ---------- is What to be displayed?

a. Window
b. all
c. Viewport
d. graphics

11- to flip the window around the x axis you set the window of dimension 640X480 to-----?

a.setwindow(0,640,0,480)
b.setwindow(640,0,480,0)
c.setwindow(0,640,480,0)
d.setwindow((640,0,0,480)

12- If the user makes the screen window bigger the previous viewport could still be viewed as previous with resizing ?

a. True
b. False

13- for the circle if Pk<0 then x remain unchanged and y is decremented by 1

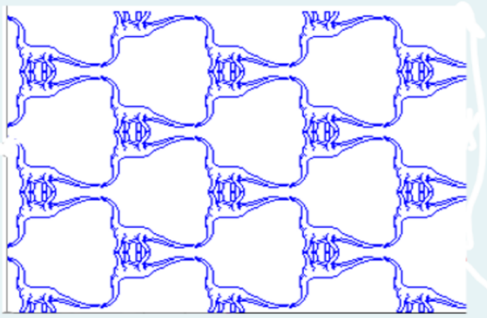
a. True
b. False

14-

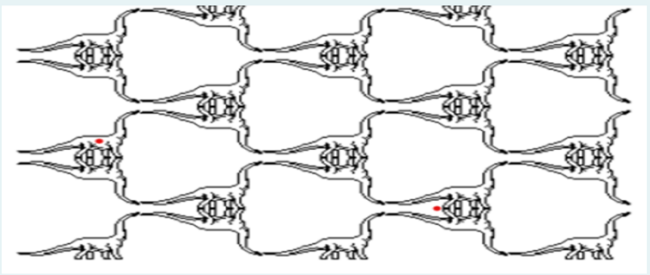
the following code draw

```
for(int i = 0; i < 5; i++)
   for(int j = 0; j < 5; j++)
   {
      if((i + j) % 2 == 0)
         setWindow(0.0, 640.0, 0.0, 480.0);
      else
         setWindow(0.0, 640.0, 480.0, 0.0);
      glViewport(i * 64, j * 44, 64, 44);
      drawPolylineFile("dino.dat");
   }
```

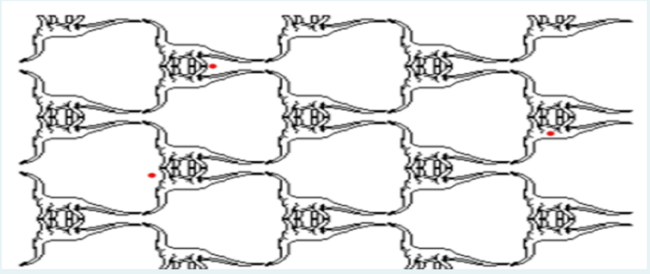
a.



b.

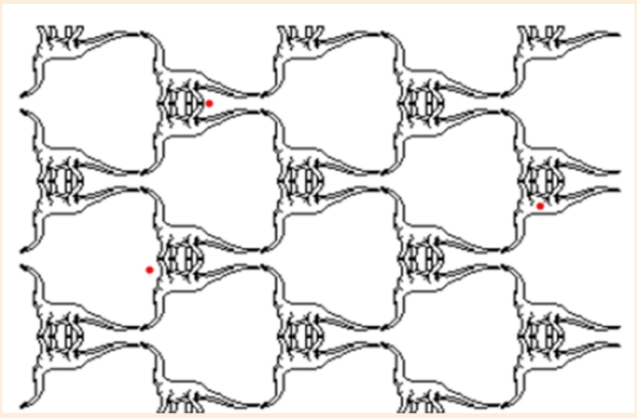


c.



Your answer is incorrect.

The correct answer is:



15-

shape B is displayed from A due to

A) 

B) 

a. the shape vertices outside the orthogonal projection
b. the shape vertices outside the window
c. the shape vertices outside the viewport
d. none

16-What is the first step in the DDA line drawing algorithm?

a. Set up the display
b. Determine the dx and dy
c. Calculate the slope
d. Initialize the drawing parameters

17-

choose the right one

offers a programmable association between pixel value and final color	Choose...
direct the electron beam gun to the location in display	Choose...
memory has the same size of the display and repeated for color depth	Choose...
It is the Cartesian coordinate which we define the diagram (XWmin, XWmax, YWmin, YWmax)	Choose...
To make the viewing process independent of the requirements of any output device, graphics systems	Choose...
The mapping of a two-dimensional, world-coordinate scene description to device coordinates is	Choose...
which convert logical values like 01 into actual voltages	Choose...
It is the screen coordinate where the objects are to be displayed (XVmin, XVmax, YVmin, YVmax)	Choose...
A section of a two-dimensional scene that is selected for display is called	Choose...

Your answer is incorrect.

The correct answer is: offers a programmable association between pixel value and final color → LUT, direct the electron beam gun to the location in display → scan controller, memory has the same size of the display and repeated for color depth → frame buffer,

It is the Cartesian coordinate which we define the diagram (XWmin, XWmax, YWmin, YWmax)
→ World coordinate,

To make the viewing process independent of the requirements of any output device, graphics systems → normalized coordinates,

The mapping of a two-dimensional, world-coordinate scene description to device coordinates is → called a two-dimensional viewing transformation, which convert logical values like 01 into actual voltages → converters,

It is the screen coordinate where the objects are to be displayed

(XVmin, XVmax, YVmin, YVmax) → Device Coordinate, A section of a two-dimensional scene that is selected for display is called → clipping window

18-

for Bresenham's-the condition

$$p_{k+1} = p_k + 2\Delta y - 2\Delta x(y_{k+1} - y_k)$$

>0 then

a. y remain unchanged
b. y change to the next point
c. the equation is valued for slope less than 1

19- Our geometric description of the "picture" will be referenced by the GLUT function------then GLUT function that Display everything and wait----------

a.glutDisplayFunc glutMainLoop ;

b.glMatrixMode gluOrtho2D (-,-,-,-)

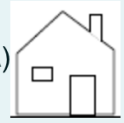
c.glBegin glEnd ;

20- _What is the initial value of the decision parameter (P) when drawing a line using Bresenham's algorithm?

a. P = Δx - Δy
b. P = 2Δy - Δx
c. P = Δy - Δx
d. P = 0

21-

to change the shape from A to B use ----------

A) 

B) 

a. Aspect ratio >1
b. Aspect ratio =1
c. Aspect ratio <1
d. change the vertices of the window

22- the condition to stope the points of the circle is--
a. the circle is drawn
b. the octane of the circle is calculated
c. all
d. x=y

23- Which algorithm may produce more rounding errors due to floating-point arithmetic?
a. Both algorithms
b. Neither algorithm
c. Bresenham's Midpoint Algorithm
d. Digital Differential Analyzer (DDA)

24- for the following setting{glOrtho2D(0,2,1,1) and glViewport(40,60,360,240)} the triangle of the vertex (10,10,10,10), (20,20,20),(30,30,30)
a. the triangle is displayed
b. nothing appeared
c. the triangle is clipped

25- A triangle is defined by three points A(2,2), B(3,5), c(4,4) what's the transformation coordinate after horizontal reflection.
a. A(-2,-2), B(-3,-5), c(-4,-4)
b. A(-2,2), B(-3,5), c(-4,4)
c. A(2,-2), B(3,-5), c(4,-4)

26- ----------is where to be displayed?
a. Window
b. all
c. graphics
d. Viewport

27- Given coordinates of window A(10,5), B(20,5), c(20,30) and D(10,30). what's the code region for the following lines. L1 ((10,30), (11,20)) L2 ((8,10), (8,20))

a. L1: 0 0 0 0 and L2 0 0 0 1
b. L1: 0 1 0 1 and L2 1 0 0 1
c. L1: 0 0 1 0 and L2 0 0 0 0

28-

```
void myReshape(GLsizei W, GLsizei H)
{
      if(R > W/H) // use (global) window aspect ratio
            setViewport(0, W, 0, W/R);
      else
            setViewport(0, H * R, 0, H);
}
```

a. keep the shape similarity with different size
b. change the window size
c. default function in OpenGL
d. change the view port size

29-

this code

```
for (int i = 0; i < num; i++)
{
  GLint x1 = random(Width);
  GLint y1 = random(Height);
  GLint x2 = random(Width);
  GLint y2 = random(Height);
  GLfloat lev = random(10)/10.0;
  glColor3f(lev,lev,lev);
  glRecti(x1, y1, x2, y2);
}
glFlush();

}|
```

generates

a. number of cubes similar is size not color
b. number of cubes
c. number of cubes identical colored similarly
d. all right

30- clipping window is the viewport?

a. True
b. False

31- How does the DDA algorithm handle cases where the slope of the line is greater than 1?

a. It uses only integer arithmetic
b. It increments the y-coordinate
c. It scales the coordinates
d. It increments the x-coordinate

Quiz 5
(trans-Q)

1- An efficient transformation method in which an object may shift form one coordinate position to any other coordinate position in a three-dimensional plane -
a. Scaling
b. Translation
c. All of the mentioned above
d. Rotation

2- The reflection around y means that all y Values--?
a. Remain unchanged
b. Become negative
c. None
d. x change to negative

3- A transformation method which may cause the change in a shape of an object is referred as,
a.Scaling
b.Rotation
c.Reflection
d.Shear

4- Amongst which one of the most basic transformation methods which may applied in three-dimensional planes -
a. Scaling
b. Translation
c. All of the mentioned above
d. Rotation

5- A line segment of a polygon may also be referred as,
a. Vertices
b. Edges
c. All of the mentioned above
d. Line

6- the object center is in the origin "In all these three transformation types, the shape of the object is never deformed."?
a. False
b. True

7- in which of the following transformation methods of Computer Graphics is the Shape of the Object not deformed?
a. Translation
b. None of the above
c. Shearing
d. Both a. and b.

8- What value of the rotation angle rotates the object in clockwise direction?
a. Value less than 180 degree
b. Positive Value
c. Negative value
d. All of the above

9- Which of the following transformation techniques is responsible for altering (either enlarging it or diminishing it) the size of the object?
a. Scaling
b. Rotation
c. Translation
d. None of the above

10- Which of the following is a matrix form of 2D translation equation is-----?
a. None of the above mentioned
b. P'= P / R
c. P'= P + R
d. P'=R*P

11- The reflection around x meanse that all X Values--?
a.Y change to negative
b.Remain unchanged
c.None
d.Become negative

12- We set all scale factors to reduce the object's dimension.
a.Less than 1
b.Greater than 0
c.in Between 0 and 1
d.None of the Above

13- An efficient transformation method which produces a parallel mirror image of an object is also referred as,
a. Shear
b. Reflection
c. Rotation
d. All of the mentioned above

14- ___ is a rigid body transformation that enables an entity to shift without deforming.?

a. Rotation
b. All
c. Translation
d. None of the above

15- What is a necessary to specify to generate rotation?

a. Repositioning
b. Scaling
c. Specific distances between dx and dy
d. Rotation angle θ

Quiz 6

Transformation, clipping, flood fill and boundary fill Quiz

1- Given coordinates of window A(10,5), B(20,5), c(20,30) and D(10,30). what's the code region for the following lines. L1 ((10,30), (11,20)) L2 ((8,10), (8,20))

a.L1: 0 0 0 0 and L2 0 0 0 1

b.L1: 0 1 0 1 and L2 1 0 0 1

c.L1: 0 0 1 0 and L2 0 0 0 0

2- Consider the following grid before the flood fill operation. If the starting pixel containing the interior color is (2, 1)(the third row and second column) and the fill color is 2, which of the following grids will result after performing a flood fill using 4-way connectivity?

0 0 0 3 3
0 1 1 3 3
0 1 1 1 3
0 0 1 3 3
0 4 4 4 3

a.

2 2 2 2 2
2 2 2 2 2
2 2 2 2 2
2 2 2 2 2
2 2 2 2 2

b.

0 0 0 3 3
0 2 2 3 3
0 2 2 2 3
0 0 2 3 3
0 4 4 4 3

c.

0 0 0 3 3
0 1 1 3 3
0 1 1 1 3
0 0 1 3 3
0 4 4 4 3

3- If we perform a flood fill on a region containing the needed interior color and use 8-way connectivity, which of the following pixels will be bushed into the stack to be filled starting from the pixel (2, 2) ?

a.All pixels that are directly connected vertically or horizontally or diagonally connected to (2, 2)
b. Only the directly adjacent diagonal pixels to (2, 2)
c. Only pixels directly connected vertically or horizontally to (2, 2)
d. No pixels will be filled as it doesn't support 8-way connectivity

4- In a flood fill algorithm, what would happen if the seed point is surrounded by the same color as the fill color?

a. The flood fill will continue to the entire grid.
b. The flood fill will only fill the pixels adjacent to the seed point.
c. The flood fill will not change any adjacent pixels.

5- In the Boundary Fill Algorithm, the color of the pixels adjacent to the starting pixel must be:

a. Different from the boundary color
b. The same as the boundary color
c. Irrelevant, as only the boundary color matters
d. The same as the fill color

6- In the boundary fill algorithm, the fill color is applied to:

a. Only diagonal pixels
b. Pixels that are of the boundary color
c. All pixels except those on the boundary

7- In the boundary fill algorithm, the fill process:

a. Always fills the entire image with a single color
b. Can only fill regions in diagonal directions
c. Uses a stack to handle the recursive function calls

8- The boundary fill algorithm in computer graphics is used to:

a. Outline the boundary of a shape
b. Apply a gradient color within a region
c. Fill a region with a specific color until a boundary is reached
d. Detect boundaries in an image

9- Which algorithm is better suited for filling regions with complex boundaries or multiple disconnected regions?

a. Both algorithms equally
b. Boundary Fill
c. Flood Fill
d. Neither algorithm

10- Which of the following defines the boundary color in the boundary fill algorithm?
a. The color of pixels that prevent further filling
b. The color of the region to be filled
c. The color of pixels connected to the fill region
d. The color of the initial fill pixel

11- Which of the following is NOT a typical condition for the boundary fill algorithm to stop?
a. When a pixel reaches a specific color (boundary color)
b. When the fill color is applied to all pixels
c. When a pixel is already filled with the specified color
d. When the algorithm reaches the edge of the image

12- Which of the following variations of flood fill allows for more flexible filling of a region, including diagonal pixels?
a. Infinite-way connectivity
b. 8-way connectivity
c. 6-way connectivity
d. 4-way connectivity

13- If both endpoints of a line have outcodes of 0000 in the Cohen-Sutherland Algorithm, what does it imply?
a. The line needs to be clipped.
b. The line is horizontal.
c. The line is completely outside the window.
d. The line is completely inside the window.

14- If one endpoint of a line has an outcode of 0010 and the other has an outcode of 0010 in the Cohen-Sutherland Algorithm, what should be done?
a. Clip the line at the window's left boundary.
b. Clip the line at the window's top boundary
c. Clip the line at the window's right boundary.
d. Both points lie outside, so no clipping is needed.

15- In the Cohen-Sutherland Algorithm, the line is classified into different regions using:
a. A 2-bit outcode.
b. A 3-bit outcode.
c. A 4-bit outcode.
d. A 5-bit outcode.

16- In Weiler-Atherton Clipping algorithm For a pair of vertices transition is outside-to-inside it follow?
a. The clipping window boundary
b. polygon boundary
c. Non

17- Rotate point p by angle 30 around point(4,3) then rotate angle by -50 around point (2,2), Which of the following transformations correctly produces the point p'?
a. P'=r(30).T(-4,-3).r(-50).T(-2,-2).p
b. P'=T(2,2).r(-50).T(2,1).r(30).T(-4,-3).p
c. P'=(30).T(-4,-3).T(4,3).r(-50).T(-2,-2).T(2,2).p

18- The following figure represents a polygon intersection with a clipping window. Applying the Hodgeman clipping algorithm the polygon is passed to one of the four clippers left, right, bottom, and top respectively, generating an output for the next clipper. What output is generated from the bottom clipper to the top clipper?
a. $v1$, $v2$, $v3$, $v3'$, $v4'$, $v5$, $v6$, $v6'$
b. $v2$, $v3$, $v3'$, $v4'$, $v5$, $v6$, $v1$
c. $v2$, $v3$, $v4$, $v5$, $v6$, $v1$

19- Which of the following bits in the Cohen-Sutherland Outcode represents the "top" boundary of the clipping window?
a. Bit 3
b. Bit 1
c. Bit 2
d. Bit 4

20- Which of the following is a correct sequence of steps in the Cohen-Sutherland Line Clipping Algorithm?
a. Only process horizontal lines.
b. Check if the line is completely outside the window then Clip the line then Accept the line.
c. Classify the line using outcodes then Check if it is inside or outside then discard, Clip or accept.
d. Discard all lines that are outside the window then Draw the lines that are inside.

Quiz 7
Quiz_Filling and perspective projection

1- A winding number of 0 means that a point is:
a. On the edge of the polygon
b. Outside the polygon
c. Inside the polygon
d. Undefined relative to the polygon

2- A larger FOV angle in perspective projection results in:
a. A more zoomed-out
b. A more zoomed-in
c. Reduced distortion in the center of the view
d. Increased distortion at the edges of the image

3- In a perspective projection matrix, which of the following parameters controls the field of view (FOV)?
a. Focal length
b. Aspect ratio
c. Near and far clipping planes
d. Camera position

4- In perspective projection, objects outside the view frustum are:
a. Clipped and not visible in the final image
b. Rendered with reduced size
c. Automatically zoomed in
d. Rendered with no distortion

5- In perspective projection, the FOV angle affects the:
a. Resolution of the rendered image
b. Size and shape of the view frustum
c. Camera's position in the scene
d. Near and far clipping planes

6- In the 3D viewing pipeline, model transformation is responsible for:
a. Changing the viewpoint and camera position
b. Mapping the 3D scene onto a 2D plane
c. Changing the position, orientation, and scale of objects in the scene
d. Applying the perspective projection

7- The 3D viewing pipeline consists of multiple stages. Which of the following is the correct sequence of these stages?
a. View transformation then Projection transformation then Model transformation then Clipping
b. Model transformation then Projection transformation then View transformation then Clipping
c. Clipping then View transformation then Projection transformation then Model transformation
d. Model transformation then View transformation then Projection transformation then Clipping

8-The near plane of the view frustum is:
a. The furthest plane from the camera
b. The closest plane to the camera
c. Invisible to the camera and only used for rendering
d. Perpendicular to the viewing direction

9- The view frustum in perspective projection is:
a. A pyramid-shaped region that represents the visible space in a 3D scene
b. A spherical region that limits the camera's perspective
c. A cylindrical region where the camera can move
d. A cube that defines the boundaries of the camera's view

10- The view frustum transformation is used in:
a. Projection transformation
b. View transformation
c. Modeling transformation
d. Lighting transformation

11- what's the Projection of the point (5,5,20) onto a screen using a camera with a FOV of 50 degrees, aspect ratio of 40:20, and near/far planes at 10 and 70 units.
a. X'=0.45, Y'=0.69
b. X'=0.27, Y'=0.54
c. X'=0.3, Y'=0.35
d. X'=0.42, Y'=0.49

12- Which of the following is the main purpose of perspective projection in computer graphics?
a. To display objects without any depth
b. To create a 2D representation of 3D objects that simulates depth
c. To scale objects to fit within the view frustum
d. To adjust the lighting and shadows in a scene

13- Which of the following is true about the vanishing point in a perspective projection?
a. It is the point where parallel lines in the scene converge
b. It is the point where all objects appear to be the same size
c. It is a fixed point on the camera's lens
d. It always lies on the z-axis in 3D space

14- If a polygon has a clockwise orientation and the point lies inside it, the winding number will be:
a. Undefined
b. 0
c. +1
d. -1

15- In the winding number method for filling all polygons, the polygon orientation has to be:
a. Both polygons have a clockwise orientation
b. Each polygon has a different orientation
c. Both polygons have a counterclockwise orientation

16- In the winding number method for filling the difference between two polygons, if a point has a positive winding number greater than 0, the point is:
a. inside one polygon but outside the other
b. Inside the intersection of the polygons
c. Inside the difference between the two polygons
d. Outside both polygons

17- In the winding number method for filling the difference between two polygons, the polygon orientation has to be:
a. Both polygons have a counterclockwise orientation
b. Both polygons have a clockwise orientation
c. Each polygon has a different orientation

18- In the winding number method for filling the intersection between two polygons, if a point has a positive winding number equal 1, the point is:
a. Inside the difference between the two polygons
b. inside one polygon but outside the other
c. Outside both polygons
d. Inside the intersection of the polygons

19- What is the first step in calculating the winding number for a point relative to a polygon?
a. Cast a ray from the point and track its intersections with the polygon edges
b. Check if the point is on the edge of the polygon
c. Calculate the area of the polygon
d. Count the number of vertices of the polygon

20- When filling all polygons using the winding number method, the point is filled if:
a. The point is on the boundary of the polygon
b. The winding number is negative
c. The winding number is nonzero
d. The winding number is zero

Quiz 8
Quiz_Makeup-1

1- By using Mid point algorithm to draw a circle with a radius of 9 units, along the circle octant in the 1st quadrant includes
a. (1,9), (2,9), (3,9), (4,9),(5,9),(8,8),(7,7)
b. (1,10), (2,9), (3,9), (4,9),(5,9),(8,8),(7,7)
c. (1,9), (2,10), (3,11), (4,11),(5,11),(8,12),(7,13)

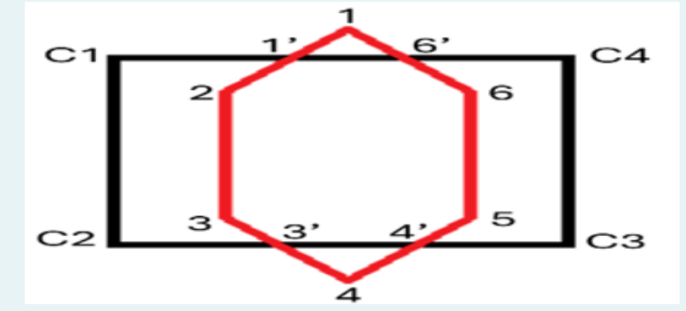
2- A triangle is defined by three points A(2,2), B(3,5), c(4,4) what's the transformation coordinate after horizontal reflection.
a. A(-2,-2), B(-3,-5), c(-4,-4)
b. A(2,-2), B(3,-5), c(4,-4)
c. A(-2,2), B(-3,5), c(-4,4)

3- If both endpoints of a line have outcodes of 0000 in the Cohen-Sutherland Algorithm, what does it imply?
a. The line is completely inside the window.
b. The line is horizontal.
c. The line is completely outside the window.
d. The line needs to be clipped.

4- If one endpoint of a line has an outcode of 0010 and the other has an outcode of 0010 in the Cohen-Sutherland Algorithm, what should be done?
a. Clip the line at the window's left boundary.
b. Clip the line at the window's top boundary
c. Clip the line at the window's right boundary.
d. Both points lie outside, so no clipping is needed.

5- In the Cohen-Sutherland Algorithm, the line is classified into different regions using:
a. A 4-bit outcode.
b. A 3-bit outcode.
c. A 5-bit outcode.
d. A 2-bit outcode.

6- The following figure represents a polygon intersection with a clipping window. Applying the Hodgeman clipping algorithm the polygon is passed to one of the four clippers left, right, bottom, and top respectively, generating an output for the next clipper. What output is generated from the bottom clipper to the top clipper?



a. $v1$, $v2$, $v3$, $v3'$, $v4'$, $v5$, $v6$, $v6'$
b. $v2$, $v3$, $v3'$, $v4'$, $v5$, $v6$, $v1$
c. $v2$, $v3$, $v4$, $v5$, $v6$, $v1$

7- Which of the following statements is true about the Weiler-Atherton Clipping algorithm?
a. The Weiler-Atherton algorithm only works for convex polygons and cannot be used for concave polygons.
b. The algorithm works by identifying the intersections of the subject polygon with the clipping region and clipping the subject polygon along the boundary of the clipping window.
c. The Weiler-Atherton algorithm is a line clipping algorithm and can only clip polygons that are fully inside or fully outside the clipping window.

8- Filled Area primitives are used to fill solid colors to a polygon; given an interior position and unified border color which of the following algorithms is best suited for polygon filling.
a. Boundary fill algorithm
b. Scan Fill Algorithm
c. Flood fill algorithm

9- In a flood fill algorithm, what would happen if the seed point is surrounded by the same color as the fill color?
a. The flood fill will not change any adjacent pixels.
b. The flood fill will only fill the pixels adjacent to the seed point.
c. The flood fill will continue to the entire grid.

10- In the Boundary Fill Algorithm, the color of the pixels adjacent to the starting pixel must be:
a. The same as the fill color
b. The same as the boundary color
c. Different from the boundary color
d. Irrelevant, as only the boundary color matters

11- In the boundary fill algorithm, the fill process:
a.Can only fill regions in diagonal directions
b.Uses a stack to handle the recursive function calls
c.Always fills the entire image with a single color

12- Which algorithm is better suited for filling regions with complex boundaries or multiple disconnected regions?
a. Neither algorithm
b. Boundary Fill
c. Flood Fill
d. Both algorithms equally

13- Which of the following variations of flood fill allows for more flexible filling of a region, including diagonal pixels?
a. Infinite-way connectivity
b. 8-way connectivity
c. 4-way connectivity
d. 6-way connectivity

14- What is a common characteristic of interactive graphics?
a. Use of traditional print media
b. Static images displayed sequentially
c. Real-time rendering and user feedback
d. Pre-rendered animations

15- Concave Shape is:
a. Polygon Shape determined if a straight line can be drawn between two points in the shape that lies outside the boundary of the shape.
b. Polygon Shape Which have number of edges equal to number of angles.
c. Polygon Shape determined if a straight line can be drawn between two points in the shape that lies inside the boundary of the shape.
d. Shape that have start point and end point not connected.

16- Graphics pipeline Defined as : It is a ..... that describes what steps a graphics system needs to perform to render a ..... to a .....
a. conceptual model - 2D scene - 3D screen
b. conceptual model - 3D scene - 2D screen
c. Design model - 3D scene - 2D screen
d. Mathematical model - 3D scene - 2D screen

17- Using the Look-Up Tables (LUTs) method for Pixel color representation, How much memory is required for a 4-bit/pixel frame buffer of size 512x1024 along with LUT of 6 byte wide?
a. 262240 byte
b. 1M byte
c. 262144 byte
d. 4K byte

18- ----------: describes how objects appears (it defines attributes that describe color texture or shape of the object
a. physical model
b. Geometric model
c. all mentioned
d. mathematical model

19- ..... Used for less details Images such as logos
a. none
b. Vector display
c. Raster display

20- DDA Line Drawing Algorithm produce the following points for a line with endpoints (5,7) and (8,14).
a. (7,11) (7,12) (7,13)
b. (5,7) (5,8) (6,10)
c. (5,9) (6,10) (6,11)

21- if the slop of line is small negative , we decide to change or not each time the value of -------
a. Y value
b. X value
c. non
d. decide each time step which to increment

22- In which algorithm is the decision parameter updated during the line drawing process?
a. Both algorithms
b. Neither algorithm
c. Bresenham's Midpoint Algorithm
d. Digital Differential Analyzer (DDA)

23- The primary advantage of Bresenham's algorithm over other line drawing algorithms is:
a. It uses floating-point arithmetic
b. It is faster and uses only integer calculations
c. It requires more memory
d. It can draw curves easily

24- Which of the following equations is used to update the decision parameter in Bresenham's algorithm if the decision parameter value was greater than zero?
a. $P_new = P + 2\Delta x$
b. $P_new = P + \Delta x - \Delta y$
c. $P_new = P + 2\Delta y - 2\Delta x$
d. $P_new = P + 2\Delta y$

25- What is the initial value of the decision parameter (P) when drawing a line using Bresenham's algorithm with endpoints (20,10) and (30, 18)?
a. P = 4
b. P = 16
c. P = 0
d. P = 6

26- In the Midpoint Ellipse Algorithm, the two axes of the ellipse are defined by which of the following parameters?
a. Major and Minor Axes
b. Height and Width
c. Radius and Diameter
d. X and Y Coordinates

27- What is the initial decision parameter used in the Midpoint Circle Drawing Algorithm for the first point of the circle?
a. p=r-2
b. p=r*r
c. p=r
d. p=1-r

28- What is the purpose of the decision parameter in the Midpoint Ellipse Algorithm?
a. To decide which pixel to plot next
b. To determine the color of the ellipse
c. To calculate the area of the ellipse
d. To set the rendering speed

29- ……to model light and to perform simulations for animation
a. mathematical model
b. all mentioned
c. physical model
d. Geometric model

Bresenham Line Drawing for a line line with endpoints (10, 0) and (20, 8). This line has a slope of 0.8, includes
a. (12,2), (13,3), (14,4), (15,4)
b. (12,2), (13,3), (14,3), (15,4)
c. (12,2), (13,2), (14,3), (15,4)